Or is it?

Theoretical discoveries made during the early decades of the twentieth century, long before string theory came on the scene, suggested that extra dimensions need not be a problem at all. And, with a late-twentieth-century updating, physicists showed that these extra dimensions have the capacity to bridge the gap between string theory's vibrational patterns and the elementary particles experimenters have discovered.

This is one of the theory's most gratifying developments; let's see how it works.

Unification in Higher Dimensions

In 1919, Einstein received a paper that could easily have been dismissed as the ravings of a crank. It was written by a little-known German mathematician named Theodor Kaluza, and in a few brief pages it laid out an approach for unifying the two forces known at the time, gravity and electromagnetism. To achieve this goal, Kaluza proposed a radical departure from something so basic, so completely taken for granted, that it seemed beyond questioning. He proposed that the universe does not have three space dimensions. Instead, Kaluza asked Einstein and the rest of the physics community to entertain the possibility that the universe has *four* space dimensions so that, together with time, it has a total of five spacetime dimensions.

First off, what in the world does that mean? Well, when we say that there are three space dimensions we mean that there are three independent directions or axes along which you can move. From your current position you can delineate these as left/right, back/forth, and up/down; in a universe with three space dimensions, any motion you undertake is some combination of motion along these three directions. Equivalently, in a universe with three space dimensions you need precisely three pieces of information to specify a location. In a city, for example, you need a building's street, its cross street, and a floor number to specify the whereabouts of a dinner party. And if you want people to show up while the food is still hot, you also need to specify a fourth